Samuel Cameron Lowe

Applied mathematician and software engineer with GIS analysis and ML modeling experience. Work recognized by NPR and the New York Times.

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Education

Northeastern University – College of Science Boston, MA
Master of Science in Applied Mathematics, Data Science Concentration, Summa Cum Laude *May 2025*

- **Relevant Coursework:** Algorithms, Machine Learning & Statistical Learning Theory 1 & 2, Probability 1 & 2, Measure Theory, Abstract Algebra 1 & 2
- **GPA:** 3.92/4.0

Northeastern University – Khoury College of Computer Sciences Boston, MA
Bachelor of Science in Computer Science and Mathematics *June 2023*

- **Relevant Coursework:** Software Development, Computer Systems, Linear Algebra, PDEs
- **Activities & Societies:** Math Club (President), Math Education Mentorship Association (Executive Board Member), Game Dev Club, NUPride

Technical Skills

Languages: Python, C#, Rust, Java, SQL, C/C++, TypeScript, Go
Technologies: Django, NumPy, Pandas, GeoPandas, Scikit-Learn, Keras, TensorFlow, Git, FastAPI, Bash, Docker, Kubernetes, Jupyter, LaTeX, Vue.js, React.js, Tableau
Domain Expertise: Data Science, Deep Learning, Data Engineering, Data Cleaning, GIS Analysis, A/B Testing, Queueing Theory, Object-Oriented, Functional, Event-Based, and Async Programming

Experience

Lead Software Engineer Boston, MA
Oasis Management Inc. *May 2026 – Present*

- Led a team of five engineers to build our enterprise resource planning website and the mobile app that automates business records, securely stores business documents, manages customer relations, and assigns work to employees
- Streamlined and automated company-wide workflows to process through months-long invoice backlog in a single day, shorten new employee onboarding from half an hour to under five minutes, and generate time cards with proof of work for every employee on the app
- Designed and built the platform end to end: ported a 15,000-line Node/Express prototype to full-stack Django/HTMX, with continuous deployment to staging and production Railway environments gated on automated tests and pre-commit hooks

Consulting Data Analyst Boston, MA
Center for Strategic Politics *Dec. 2024 – Present*

- Designed the weighted single transferable vote algorithm to simulate the outcomes for the NYC Mayoral Democratic Primary Poll, which was featured on the New York Times
- Conducted statistical poll analysis, and performed geographic information system analysis that has been commissioned by state-level and congressional campaigns around the country

Software Engineering Co-op Boston, MA
BitSight Technologies *Jan. 2022 – Aug. 2022*

- Designed and prototyped Golang microservice to parallelize data processing

- Added features and unit tests to a Django back end with SQL, Python, and Kubernetes

Back-End Engineer Boston, MA
Generate Product Development *Jan. 2021 – May 2021*

- Worked with a team of two other back-end engineers to integrate Google Maps, RapidSOS, Bluetooth, and MySQL into our Django back-end
- Led several weekly standups to discuss team progress, priorities, and resolve issues

Projects | samlowe.dev/portfolio

nprcore | Vue.js, JavaScript, Spotify API, Git | Repository

- Pair-programmed a website to analyze users' music choices and compare them with NPR's recommended songs, which was then recognized by official NPR Twitter accounts including All Things Considered and NPR Interns
- Designed algorithms to score user music tastes against population data

Pisa | Python, Django, Lean4, SQL, JavaScript, HTML, Docker | Repository

- Built a full-stack proof assignment platform in Django for instructors to design, assign, and automatically grade assignments in the Lean4 programming language
- Features include homework and problem design, viewing and exporting student grades across multiple classes, and code environment with syntax highlighting and server-side automated feedback and grading

Gems in the Rough | Python, PySide6, Hugging Face, Gemma4 | Repository

- Cross-platform, AI-powered image sorting utility that asks Google DeepMind's Gemma4 artificial intelligence which directories images should be sorted into
- Streams images and AI suggestions to a resizable window where the user accepts or rejects each suggestion with hotkeys or buttons, moving accepted images automatically
- Multi-threading architecture allows multiple AI agents to sort images in advance of the user, providing a quick and seamless user experience

League of Losers | Python, Jupyter, NumPy, TensorFlow, scikit-learn, Matplotlib | Writeup, Repository

- Explored a data set of League of Legends team winrates based on the 10-minute scoreboard
- Trained logistic regression, random forest, neural network, linear and quadratic discriminant analysis, k-nearest neighbors, and other classifiers to predict outcomes with over 70% accuracy
- Compiled an extensive writeup detailing data cleaning, dimensionality reduction using singular value decomposition, and contrasted the performance of various models

PaperScraper | Python, PyTest, Poetry, uv, APIs (Reddit, Imgur, Flickr), Git | Writeup, Repository

- Developed a command-line program to allow users to easily download wallpapers from Reddit
- Used asynchronous programming to improve performance when downloading multiple files
- Wrote automated unit tests with PyTest, MagicMocks, VCR.py, GitHub Actions